Chapter 3

The PN Junction Diode

3.1 Introduction

In the previous lecture, we studied N-type and P-type semiconductors individually. While they conduct electricity better than intrinsic silicon, they act largely as simple resistors. The true "magic" of electronics happens when we physically join a P-type material to an N-type material.

This boundary is called the **PN Junction**. This single structure is the fundamental building block of almost all modern semiconductor devices, including the **Diode, the Transistor (BJT & FET), the LED, and the Solar Cell.**

Learning Objectives

By the end of this chapter, you should be able to:

- Explain the processes of **Diffusion** and **Recombination**.
- Describe the formation of the **Depletion Region** and the **Barrier Potential**.
- Analyze the behavior of the PN junction under **Forward** and **Reverse Bias**.
- Sketch the V-I characteristic curve of a practical diode.

3.2 Formation of the PN Junction

Imagine taking a single crystal of Silicon and doping one side with Boron (Acceptors) to create P-type material, and the other side with Phosphorus (Donors) to create N-type material.

3.2.1 The Diffusion Process

At the instant the junction is formed:

1. The **N-region** has a very high concentration of Free Electrons.
2. The **P-region** has a very high concentration of Holes.
3. Due to the concentration gradient, charges naturally move from areas of high concentration to low concentration. This is called **Diffusion**.
4. Free Electrons from the N-side cross the junction into the P-side.
5. Holes from the P-side cross into the N-side.

3.2.2 Recombination

When a Free Electron crosses into the P-region, it is surrounded by Holes. It immediately falls into a hole. This process is called **Recombination**.

- The free electron vanishes (becomes a bound valence electron).
- The hole vanishes (it is filled).

3.3 The Depletion Region

The recombination process has a permanent side effect near the junction boundary.

3.3.1 Formation of Ions

- When an N-type atom loses its free electron, it leaves behind a net positive charge in its nucleus. It becomes a **Positive Ion**.
- When a P-type atom gains an electron (filling its hole), it acquires a net negative charge. It becomes a **Negative Ion**.

Unlike free carriers, these ions are locked in the crystal lattice and **cannot move**.

3.3.2 The Wall

This creates a region near the junction containing:

- Fixed **Negative Ions** on the P-side.
- Fixed **Positive Ions** on the N-side.
- **NO Free Carriers** (they have all recombined).

This region is depleted of charge carriers and is therefore called the **Depletion Region**.

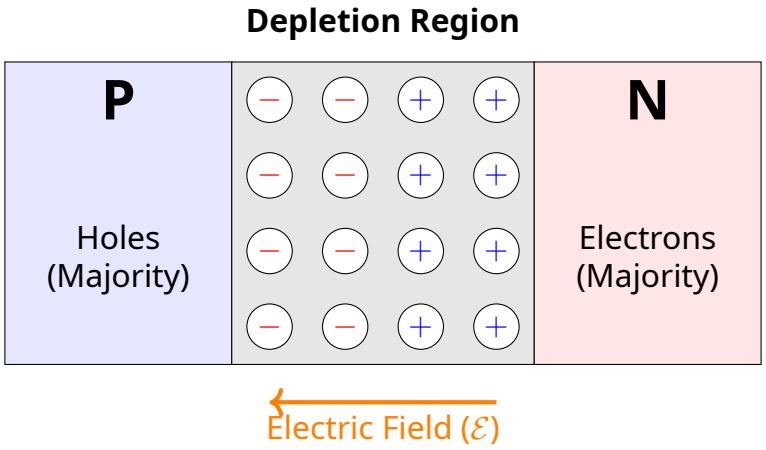


Figure 3.1: The Depletion Region showing immobile ions and the resulting Electric Field.

3.4 Barrier Potential (V_0)

The rows of positive and negative ions create an internal electric field ($\mathcal{E}$). This field acts as a force that opposes further diffusion.

- An electron trying to cross from N to P is repelled by the negative ions on the P-side.
- Eventually, equilibrium is reached. Diffusion stops.

The potential difference created across the depletion region is called the **Barrier Potential (V_0)**. To get current to flow through the device, we must apply an external voltage greater than this barrier.

Standard Barrier Potentials

At room temperature ($25°$C):

- **Silicon (Si):** $V_0 \approx 0.7$ V
- **Germanium (Ge):** $V_0 \approx 0.3$ V

3.5 Biasing the PN Junction

"Biasing" means applying an external DC voltage to a semiconductor device to control its operation.

3.5.1 Forward Bias

To make the diode conduct, we must push carriers toward the junction to overcome the depletion region.

- Connect **Positive** terminal to **P-type** (Anode).
- Connect **Negative** terminal to **N-type** (Cathode).

Mechanism:

1. The positive terminal repels holes in the P-region toward the junction.
2. The negative terminal repels electrons in the N-region toward the junction.
3. The depletion width is compressed and narrows.
4. If $V_{applied} > 0.7$V, the barrier collapses, and heavy current flows.

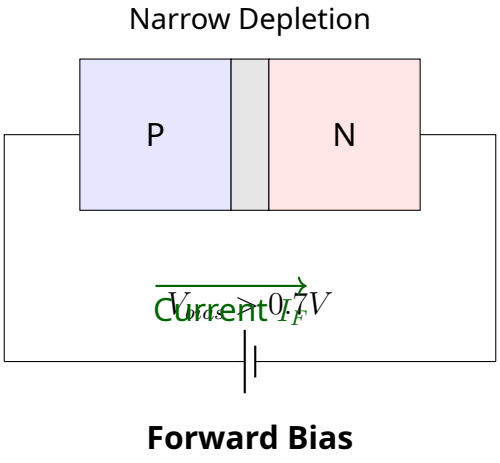


Figure 3.2: Forward Bias Configuration. The potential barrier is reduced.

3.5.2 Reverse Bias

To block current, we pull carriers away from the junction.

- Connect **Negative** terminal to **P-type**.
- Connect **Positive** terminal to **N-type**.

Mechanism:

1. The positive terminal attracts N-electrons *away* from the junction.
2. The negative terminal attracts P-holes *away* from the junction.
3. The depletion width **widens**.
4. The barrier potential increases. No majority current can flow. The diode acts as an Open Switch.

3.6 3.6 Diode Symbol and Terminals

The schematic symbol for a diode is an arrow head pointing to a line.

- The arrow represents the **Anode** (P-Type).
- The line represents the **Cathode** (N-Type).
- The arrow points in the direction of conventional current flow.

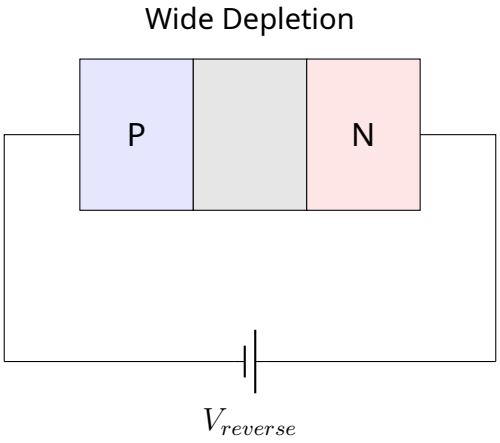


Figure 3.3: Reverse Bias Configuration. The potential barrier is increased.

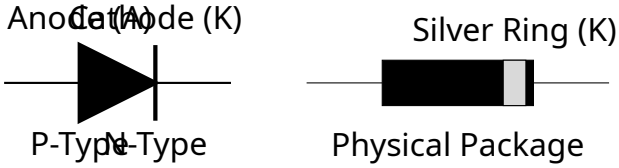


Figure 3.4: Diode Circuit Symbol and Physical Package.

3.7 Review Questions

Part A: Conceptual Questions

1. Define ”Diffusion” in the context of semiconductors. What causes it?
2. Explain why the Depletion Region acts as an insulator.
3. What is the approximate Barrier Potential for a Silicon diode and a Germanium diode?
4. Draw the circuit symbol for a diode and label the Anode and Cathode.
5. Explain what happens to the width of the depletion region when a diode is:
   (a) Forward Biased
   (b) Reverse Biased
6. What is the purpose of the ”Silver Ring” on a physical diode package?
7. Why does current flow in a diode only in one direction?

8. Define "Reverse Saturation Current" (I_S). What causes it?

9. What is "Recombination" and where does it occur?

10. In a forward-biased diode, do electrons flow with or against the direction of the arrow symbol?

Part B: Advanced Calculations & Analysis

11. **Built-in Potential Calculation:** The built-in potential barrier V_0 is given by the equation:
$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$
Calculate V_0 for a Silicon PN junction at 300 K if doping concentrations are $N_A = 10^{17}$ cm^{-3} and $N_D = 10^{16}$ cm^{-3}. Assume $n_i = 1.5 \times 10^{10}$ cm^{-3} and $V_T = 26$ mV.

12. **Temperature Sensitivity:** The reverse saturation current (I_S) of a diode doubles for every 10°C rise in temperature. If $I_S = 5$ nA at 25°C, calculate the leakage current at 75°C.

13. **Shockley Equation Application:** The current through a diode is given by $I_D = I_S(e^{V_D/nV_T} - 1)$. Calculate the diode current if $I_S = 10^{-12}$ A, $V_D = 0.65$ V, $n = 1$, and $V_T = 26$ mV.

14. **Depletion Width Analysis:** The width of the depletion region (W) is proportional to $\sqrt{V_0 + V_R}$, where V_R is the applied reverse voltage. If the width is 1μm at $V_R = 3$V (assume $V_0 \approx 0.7$V), estimate the new width if the reverse voltage is increased to 10V.

15. **Dynamic Resistance:** The AC resistance of a diode is given by $r_d = \frac{26\text{ mV}}{I_D}$. If a diode is biased at 2 mA, calculate its dynamic resistance. How does this resistance change if the current is increased to 10 mA?